1. find - Search for Files

Find all log files:

find /var/log -type f -name "*.log"

Find files larger than 100 MB:

find /var -type f -size +100M

Explanation:

- -type f: Search only regular files.
- -name: Match filenames.
- -size +100M: Find files larger than 100 MB.

2. grep - Search Text in Files

grep -rin "error" /var/log/

Explanation:

- -r: Search recursively.
- -i: Ignore case.
- -n: Display line numbers.

Search failed SSH logins:

sudo grep -i "failed password" /var/log/secure

3. awk - Extract Columns from Files

awk -F: '{print $1, $3}' /etc/passwd

Explanation:

- -F:: Use colon as the field separator.
- $1: First field (username).
- $3: Third field (user ID).

Use case: Extracting information from configuration files and command output.

4. sed - Replace Text

Display replaced text without modifying the file:

sed 's/oldserver/newserver/g' config.txt

Modify the file directly:

sed -i 's/oldserver/newserver/g' config.txt

Explanation:

- s: Substitute text.
- g: Replace all occurrences on each line.
- -i: Edit the file directly.

Warning: Back up important files before using -i.

5. xargs - Pass Output to Another Command

find /tmp -type f -name "*.tmp" -print0 | xargs -0 ls -lh

Explanation:

- find: Locates temporary files.
- -print0: Separates filenames safely.
- xargs -0: Passes filenames to another command.
- ls -lh: Displays file details in human-readable format.

Use case: Processing many files efficiently.

6. tar - Archive and Compress Directories

Create a compressed backup:

tar -czvf backup.tar.gz /home/user/project

Extract the archive:

tar -xzvf backup.tar.gz

Explanation:

- -c: Create an archive.
- -x: Extract an archive.
- -z: Use gzip compression.
- -v: Display processed files.
- -f: Specify the archive filename.

7. lsof - Identify Processes Using Files or Ports

Check which process uses port 8080:

sudo lsof -i :8080

Check which processes have a file open:

sudo lsof /var/log/messages

Explanation:

- lsof means list open files.
- -i :8080: Display network activity associated with port 8080.

If missing on SUSE:

sudo zypper install lsof

8. ss - Check Network Ports

Display listening TCP and UDP ports:

sudo ss -tulpn

Check port 8080:

sudo ss -ltnp 'sport = :8080'

Explanation:

- -t: TCP sockets.
- -u: UDP sockets.
- -l: Listening sockets.
- -p: Associated processes, when permitted.
- -n: Numeric addresses and ports.

9. systemctl - Manage System Services

Check Jenkins status:

sudo systemctl status jenkins

Start Jenkins:

sudo systemctl start jenkins

Stop Jenkins:

sudo systemctl stop jenkins

Restart Jenkins:

sudo systemctl restart jenkins

Enable Jenkins at startup:

sudo systemctl enable jenkins

Explanation:
systemctl manages services on Linux systems that use systemd.

10. journalctl - View System Logs

View Jenkins logs:

sudo journalctl -u jenkins

Display the last 50 Jenkins log entries:

sudo journalctl -u jenkins -n 50

Follow Jenkins logs live:

sudo journalctl -u jenkins -f

View logs from the past hour:

sudo journalctl --since "1 hour ago"

Explanation:

- -u: Filter by service unit.
- -n 50: Display the last 50 entries.
- -f: Follow new log entries live.
- --since: Filter by time.